Banach Tarski Paradox

Mohammad Javad Moghaddas Mehr*

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Abstract

This paper gives a self-contained proof of the Banach–Tarski paradox. We outline the foundational context, develop the required group-theoretic machinery, construct a paradoxical decomposition of the sphere, and extend it to a solid ball using equidecomposability. The proof illustrates how free group actions and the Axiom of Choice combine to produce the classical paradox.

Keywords: Banach–Tarski Paradox, Axiom of Choice, Measure Theory, Paradoxical Decomposition

1 Introduction

The problem of assigning a meaningful notion of "size" to arbitrary subsets of $\mathbb{R}^n$ lies at the heart of measure theory and modern analysis. Classical intuition suggests that such a notion should extend the familiar geometric concepts of length, area, and volume, and should respect the underlying symmetries of Euclidean space. One is therefore naturally led to seek a measure μ defined on all subsets of $\mathbb{R}^n$ that satisfies the following basic requirements:

1. *countable additivity* on disjoint families;
2. *invariance* under rigid motions;
3. *normalization* on standard sets, such as $\mu([0,1]^n) = 1$.

A classical argument shows that these requirements are incompatible. By using the axiom of choice to select one representative from each orbit under rational translation, one

*m.moghadas11235@gmail.com

can partition the unit interval in a way that forces any countably additive, translation-invariant measure to assign contradictory values. Thus no such measure can exist on all subsets of $\mathbb{R}$.

In higher dimensions an even more striking phenomenon occurs. In 1924 Banach and Tarski [1] showed that in $\mathbb{R}^n$ for $n \geq 3$, one may dispense with countable additivity entirely and still encounter impossibility. They proved that any two bounded open sets in $\mathbb{R}^3$ are *equidecomposable*: each can be partitioned into finitely many pieces that can be rearranged by rigid motions to form the other. In particular, a solid ball can be divided into finitely many disjoint, highly nonmeasurable pieces, which may then be reassembled into two balls congruent to the original. A sharp result due to Robinson [3] shows that such a doubling of the ball can be accomplished with *exactly* five, and fewer than five pieces will not suffice.

At the conceptual core of the Banach–Tarski paradox is the idea of a *paradoxical decomposition*: a set that can be partitioned into finitely many pieces which, after applying suitable isometries, form two disjoint copies of the original set. Tarski's alternative states that a nonempty G-set is paradoxical precisely when it admits no G-invariant finitely additive probability measure defined on all its subsets [2]. A free action of a nonabelian free subgroup provides the mechanism used here; the presence of such a subgroup alone does not make every action paradoxical. This connection places group theory—particularly the behavior of free groups and their actions—at the center of the phenomenon. The classical proof constructs a paradoxical decomposition of the free group F_2, embeds F_2 into $\mathrm{SO}(3)$, and uses this embedding to transfer the paradox to the unit sphere S^2. A radial extension then propagates the decomposition to the entire unit ball.

The purpose of this report is to give a streamlined, self-contained exposition of this strategy. In Section 2 we review the relevant concepts from group actions, equidecomposability, and paradoxical decompositions. Section 3 develops the proof: we obtain a paradoxical decomposition of the free group F_2, embed it into $\mathrm{SO}(3)$, transfer the resulting decomposition to S^2, and extend it to the solid unit ball via radial invariance. The presentation emphasizes the interplay between group-theoretic structure and the axiom of choice, showing how these ingredients combine to yield one of the most remarkable results in modern set theory and geometric analysis.

2 Preliminaries

In this section we introduce the basic notions required for the Banach–Tarski paradox. Throughout, G denotes a group and X a set. Our goal is to understand how group actions allow us to "move pieces around," and how such motions lead to equidecompositions and paradoxical decompositions.

Definition 2.1 (Group Action)**.** *A* (left) action *of G on X is a function*

$$G \times X \longrightarrow X, \qquad (g, x) \mapsto g \cdot x,$$

such that for all $g, h \in G$ and all $x \in X$,

1. $e \cdot x = x$, *where* e *is the identity element of* G;

2. $(gh) \cdot x = g \cdot (h \cdot x)$.

In this situation we say that G acts *on* X.

Remark 2.1. *A* right *action is defined similarly using a function* $X \times G \to X$ *and the rule* $(x \cdot g) \cdot h = x \cdot (gh)$.

Definition 2.2 (Equidecomposability)**.** *Let* G *act on a set* X. *Two subsets* $A, B \subseteq X$ *are* G-equidecomposable *if there exist pairwise disjoint subsets* $A_1, \ldots, A_n \subseteq A$ *and elements* $g_1, \ldots, g_n \in G$ *such that*

$$A = \bigsqcup_{i=1}^{n} A_i, \qquad B = \bigsqcup_{i=1}^{n} g_i \cdot A_i.$$

In this case we write $A \sim_G B$ *and say that* A *can be partitioned into finitely many pieces and moved by elements of* G *to form* B.

Definition 2.3 (Paradoxical Decomposition)**.** *Let* G *act on* X. *A subset* $A \subseteq X$ *admits a* paradoxical decomposition *if there exist pairwise disjoint subsets*

$$A_1, \ldots, A_m, \qquad B_1, \ldots, B_n \subseteq A$$

and elements

$$g_1, \ldots, g_m, \qquad h_1, \ldots, h_n \in G$$

such that

$$A = \bigcup_{i=1}^{m} g_i \cdot A_i \qquad \text{and} \qquad A = \bigcup_{j=1}^{n} h_j \cdot B_j.$$

That is, A *can be partitioned into finitely many pieces which, after applying elements of* G, *form two disjoint copies of* A.

Lemma 2.1. *If* $A \sim_G B$ *and* A *admits a paradoxical decomposition under* G, *then* B *also admits a paradoxical decomposition under* G.

Proof. Let $P_i, Q_j \subseteq A$ be the mutually disjoint source pieces of a paradoxical decomposition. If images overlap within either reassembly, assign each point to its first image piece and restrict the source pieces to the corresponding preimages. Thus we may assume

$$A = \bigsqcup_{i} g_i P_i = \bigsqcup_{j} h_j Q_j.$$

Write the given equidecomposition as

$$A = \bigsqcup_{k=1}^{r} C_k, \qquad B = \bigsqcup_{k=1}^{r} t_k C_k.$$

For each i, k, ℓ, form the source piece in B

$$P'_{ik\ell} = t_k\big(C_k \cap P_i \cap g_i^{-1} C_\ell\big).$$

Move it by $t_\ell g_i t_k^{-1}$. Its image is $t_\ell(g_i(C_k \cap P_i) \cap C_\ell)$, and these images partition B. Likewise, the pieces

$$Q'_{jk\ell} = t_k\big(C_k \cap Q_j \cap h_j^{-1} C_\ell\big)$$

move by $t_\ell h_j t_k^{-1}$ to give another partition of B. All the P' and Q' source pieces are mutually disjoint: within each $t_k C_k$, the intersections refine the mutually disjoint original pieces, and the $t_k C_k$ themselves are disjoint. There are only finitely many pieces, so this is a paradoxical decomposition of B. ∎

Lemma 2.2 (Absorption by disjoint translates)**.** *Let G act on X, and let $D \subseteq A \subseteq X$. Suppose there exists $g \in G$ such that the sets $g^n D$, for $n = 0, 1, 2, \ldots$, are pairwise disjoint and all lie in A. Then $A \sim_G A \setminus D$.*

Proof. Set

$$C = \bigcup_{n=0}^{\infty} g^n D \subseteq A.$$

The disjointness assumption gives $gC = C \setminus D$. Partition A into C and $A \setminus C$. Apply g to the first piece and the identity to the second. Their images form the disjoint union

$$gC \sqcup (A \setminus C) = A \setminus D,$$

which proves the claimed two-piece equidecomposition. ∎

A paradoxical decomposition is the formal mechanism behind the Banach–Tarski paradox. In Theorem 3.2 we will show that the closed unit ball $B = \{x \in \mathbb{R}^3 : \|x\| \leq 1\}$ admits a paradoxical decomposition under the group of orientation-preserving rigid motions of $\mathbb{R}^3$. In particular, B is equidecomposable with two disjoint copies of itself.

3 The Banach–Tarski Paradox

In this section we give a detailed proof of the Banach–Tarski paradox following the classical strategy:

1. construct a paradoxical decomposition of F_2; And realize this free group as a subgroup of the rotation group SO(3);
2. transfer the paradoxical decomposition from F_2 to the unit sphere S^2;
3. extend the paradoxical decomposition of S^2 to the solid unit ball in $\mathbb{R}^3$.

Throughout, G-equidecomposability and paradoxical decomposition are meant in the sense introduced in Section 2.

3.1 A paradoxical decomposition of the free group F_2

Let F_2 be the free group on generators a and b, equipped with its standard description in terms of reduced words.

Definition 3.1. *A word $w = x_1 x_2 \cdots x_n$ in the alphabet $\{a, a^{-1}, b, b^{-1}\}$ is* reduced *if no letter is adjacent to its inverse. The free group F_2 is the set of reduced words together with the empty word e, with multiplication given by concatenation followed by cancellation of all occurrences of xx^{-1} and $x^{-1}x$.*

Every nontrivial element of F_2 begins with exactly one of the letters a, a^{-1}, b, or b^{-1}. Define the corresponding subsets:

$$\begin{aligned} S(a) &= \{\, w \neq e : w \text{ begins with } a \,\}, \\ S(a^{-1}) &= \{\, w \neq e : w \text{ begins with } a^{-1} \,\}, \\ S(b) &= \{\, w \neq e : w \text{ begins with } b \,\}, \\ S(b^{-1}) &= \{\, w \neq e : w \text{ begins with } b^{-1} \,\}. \end{aligned}$$

These four sets are disjoint and partition all nontrivial words, giving

$$F_2 = \{e\} \cup S(a) \cup S(a^{-1}) \cup S(b) \cup S(b^{-1}). \tag{1}$$

For any $w \in S(a^{-1})$ we may write $w = a^{-1}u$, so $aw = aa^{-1}u \to u$ after reduction. Conversely, every reduced word not beginning with a arises uniquely in this way. Applying the same reasoning for b, we obtain two disjoint decompositions:

$$F_2 = S(a) \cup aS(a^{-1}), \qquad \text{and} \qquad F_2 = S(b) \cup bS(b^{-1}).$$

Each of these equalities expresses F_2 as a finite union of disjoint pieces that are moved by a single group element (namely a or b) to reassemble the whole group. Taken together, they give two different reassemblies of F_2 from finitely many translated pieces.

Corollary 3.1. *The free group F_2, acting on itself by left translation, admits a paradoxical decomposition. In particular, F_2 is F_2-paradoxical.*

Theorem 3.1. *There exist rotations $A, B \in \mathrm{SO}(3)$ such that the subgroup $H = \langle A, B \rangle$ is isomorphic to F_2.*

Proof. Take rotations by $\theta = \arccos(1/3)$ about the z- and x-axes:

$$A = \frac{1}{3}\begin{pmatrix} 1 & -2\sqrt{2} & 0 \\ 2\sqrt{2} & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad B = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & -2\sqrt{2} \\ 0 & 2\sqrt{2} & 1 \end{pmatrix}.$$

Both matrices are orthogonal with determinant one. We show that every nonempty reduced

word in $A^{\pm 1}, B^{\pm 1}$ differs from the identity.

Work in the field $K = \mathbb{F}_3[t]/(t^2+1)$, and write η for the class of t. This is a field because t^2+1 has no root in $\mathbb{F}_3$; moreover, $\eta^2 = -1 = 2$ and $\eta \neq 0$. Reducing integer coefficients modulo three and sending $\sqrt{2}$ to η defines a ring homomorphism $\mathbb{Z}[\sqrt{2}] \to K$. For $\varepsilon \in \{1, -1\}$, let $N_{A,\varepsilon}$ and $N_{B,\varepsilon}$ be the images of $3A^{\varepsilon}$ and $3B^{\varepsilon}$ under this homomorphism. They factor as

$$N_{A,\varepsilon} = \begin{pmatrix} 1 \\ -\varepsilon\eta \\ 0 \end{pmatrix} \begin{pmatrix} 1 & \varepsilon\eta & 0 \end{pmatrix} = u_{A,\varepsilon} v_{A,\varepsilon},$$

$$N_{B,\varepsilon} = \begin{pmatrix} 0 \\ 1 \\ -\varepsilon\eta \end{pmatrix} \begin{pmatrix} 0 & 1 & \varepsilon\eta \end{pmatrix} = u_{B,\varepsilon} v_{B,\varepsilon}.$$

The row-column products between adjacent factors are

$$v_{A,\varepsilon} u_{A,\delta} = v_{B,\varepsilon} u_{B,\delta} = 1 + \varepsilon\delta,$$
$$v_{A,\varepsilon} u_{B,\delta} = \varepsilon\eta, \qquad v_{B,\varepsilon} u_{A,\delta} = -\delta\eta.$$

For equal generators, this scalar vanishes exactly when the adjacent letters are inverses; for different generators, it never vanishes. Consequently, the product of the N matrices associated with a reduced word is a nonzero scalar times the outer product of its first column factor and last row factor, and hence is nonzero.

If a reduced word $W = L_1 \cdots L_n$, $n \geq 1$, were the identity, then

$$(3L_1) \cdots (3L_n) = 3^n I.$$

Reduction to K would make the right side zero and the left side nonzero, a contradiction. Thus the homomorphism from F_2 sending its generators to A, B is injective, and its image is $\langle A, B \rangle$. ■

We henceforth identify F_2 with its copy $H = \langle A, B \rangle \subset \mathrm{SO}(3)$.

3.2 A paradoxical decomposition of the sphere

Let S^2 be the unit sphere. The action of H on S^2 is not free everywhere: rotations have two fixed points each.

Lemma 3.1. *Let $D \subset S^2$ be the set of points fixed by some nontrivial element of H. Then D is countable.*

Proof. Every nontrivial rotation fixes exactly two antipodal points, and H is countable. Thus D is a countable union of two-point sets. ■

Let $S_0^2 := S^2 \setminus D$. The action of H on S_0^2 is free. By the axiom of choice, we may select a set $M \subset S_0^2$ containing exactly one representative from each H-orbit. The map

$$H \times M \to S_0^2, \qquad (h, m) \mapsto h \cdot m$$

is bijective. Let

$$H = \bigcup_{i=1}^{k} E_i$$

be the finite partition of H. For each i define

$$E_i \cdot M := \{h \cdot m : h \in E_i,\ m \in M\}.$$

Then sets $E_i \cdot M$ form a partition of S_0^2.

Proposition 3.1. *S_0^2 admits a paradoxical decomposition under the action of H.*

Proof. Apply the paradoxical decomposition of H and transfer each piece E_i to $E_i \cdot M$. The relations are preserved because the action of H on S_0^2 is free. ■

Proposition 3.2. *S_0^2 admits a paradoxical decomposition under the action of H.*

Proof. Transfer the paradoxical decomposition of H to S_0^2 using the bijection $H \times M \to S_0^2$. We obtain disjoint families

$$A_i \cdot M, \qquad B_j \cdot M$$

whose translates satisfy

$$S_0^2 = \bigcup_{i=1}^{m} g_i(A_i \cdot M) \qquad \text{and} \qquad S_0^2 = \bigcup_{j=1}^{n} h_j(B_j \cdot M),$$

which is a paradoxical decomposition of S_0^2. ■

To restore the omitted pole points, we enlarge the permitted group from H to $\mathrm{SO}(3)$ and construct a rotation satisfying the absorption lemma.

Lemma 3.2. *S^2 and S_0^2 are $\mathrm{SO}(3)$-equidecomposable.*

Proof. Since D is countable, choose an axis through the origin whose two poles are outside D. Let ρ_θ denote rotation by angle θ about this axis. For fixed $d, d' \in D$ and a positive integer k, the equation $\rho_\theta^k d = d'$ excludes only finitely many angles in $[0, 2\pi)$: neither point lies on the axis, so at most one angle modulo 2π sends d to d', and it has at most k preimages under multiplication by k. There are countably many triples (d, d', k), so we may choose an angle avoiding all these equations. Write ρ for the resulting rotation.

The sets $\rho^n D$, $n \geq 0$, are pairwise disjoint. Indeed, an intersection $\rho^n d = \rho^m d'$ with $n > m$ would give $\rho^{n-m} d = d'$, contrary to our choice. They all lie in S^2, so Lemma 2.2, with $A = S^2$ and $g = \rho$, gives

$$S^2 \sim_{\mathrm{SO}(3)} S^2 \setminus D = S_0^2.$$

■

Corollary 3.2. *The unit sphere S^2 admits a paradoxical decomposition under rotations.*

Proof. The H-paradoxical decomposition of S_0^2 is also a SO(3)-paradoxical decomposition, since $H \subseteq \mathrm{SO}(3)$. The preceding lemma and symmetry of equidecomposability give $S_0^2 \sim_{\mathrm{SO}(3)} S^2$. Apply Lemma 2.1. ■

3.3 From the sphere to the ball

We now extend the paradoxical decomposition of the sphere to the closed unit ball $B = \{x \in \mathbb{R}^3 : \|x\| \leq 1\}$.

Lemma 3.3. *If $S^2 = \bigcup A_i$ is paradoxical under rotations, then the sets*

$$\widetilde{A}_i := \{rx : x \in A_i,\ 0 < r \leq 1\}$$

form a paradoxical decomposition of $B \setminus \{0\}$.

Proof. Every point of $B \setminus \{0\}$ has a unique radial representation $x = ru$ with $u \in S^2$ and $0 < r \leq 1$. Rotations preserve radial lines: $R(rx) = r\,R(x)$. Hence any paradoxical decomposition of S^2 lifts along radial segments to a paradoxical decomposition of $B\setminus\{0\}$. ■

Finally we reincorporate the center point. For this step, let $\mathcal{G}$ be the group of orientation-preserving rigid motions of $\mathbb{R}^3$. It contains SO(3) and also rotations about axes away from the origin.

Lemma 3.4. *B and $B \setminus \{0\}$ are $\mathcal{G}$-equidecomposable.*

Proof. Consider the circle

$$\Gamma = \left\{ \left(\tfrac{1}{4} + \tfrac{1}{4}\cos t, \tfrac{1}{4}\sin t, 0\right) : t \in \mathbb{R} \right\}.$$

It passes through the origin and lies entirely in B, since each of its points has norm at most $1/2$. Let $\sigma \in \mathcal{G}$ be rotation by $2\pi\alpha$, where α is irrational, about the line $\{(1/4, 0, z) : z \in \mathbb{R}\}$ through the center of this circle. Then σ preserves Γ, and the points $\sigma^n(0)$, $n \geq 0$, are distinct: an equality would imply $(n-m)\alpha \in \mathbb{Z}$ for $n \neq m$. Thus the sets $\sigma^n\{0\}$ are pairwise disjoint and all lie in B. Lemma 2.2 gives

$$B \sim_{\mathcal{G}} B \setminus \{0\}.$$

■

Theorem 3.2 (Banach–Tarski Paradox)**.** *The closed unit ball $B \subset \mathbb{R}^3$ admits a paradoxical decomposition under orientation-preserving rigid motions. In particular, B is equidecomposable with two disjoint copies of itself.*

Proof. By Corollary 3.2, the sphere S^2 is SO(3)-paradoxical. Radial extension gives a SO(3)-paradoxical decomposition of $B \setminus \{0\}$, which is also $\mathcal{G}$-paradoxical. By Lemma 3.4 and symmetry, $B \setminus \{0\} \sim_{\mathcal{G}} B$. Lemma 2.1 therefore makes B $\mathcal{G}$-paradoxical. Finally, translate the two reassembled copies far enough apart to obtain two disjoint balls congruent to B. ■

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